

# JISHNU EC

SOC Analyst | Penetration Tester | VAPT Specialist

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## EDUCATION

Bachelor of Computer Applications (BCA)

2022 – 2025

Calicut University, Kerala

## PROFESSIONAL SUMMARY

Offensive security practitioner and BCA graduate with hands-on capability across the full red team kill chain — active reconnaissance, vulnerability exploitation, post-exploitation, lateral movement, and persistence — paired with blue team proficiency in SIEM-driven threat detection, log forensics, and structured incident response. Holder of the **EC-Council Certified SOC Analyst (CSA)** credential. Executed an independent VAPT engagement against Calicut University's web infrastructure, identifying injection vectors, broken authentication controls, and server-side misconfiguration. Built and shipped production-grade security tooling in Python and Go: a modular reconnaissance engine, a full-featured OSINT framework with automated reporting, and a publicly distributed CTF machine simulating realistic Linux privilege escalation chains. Experienced in **static and dynamic reverse engineering** — PE binary analysis, disassembly with Ghidra, behavioral IOC extraction, and YARA rule authoring. Equally at home writing an exploit, dissecting a malicious binary, or building detection logic in Splunk to catch it. Targeting a **VAPT or SOC Analyst** position where technical depth drives real security outcomes.

## CERTIFICATIONS

Certified SOC Analyst (CSA) | EC-Council | Completed 2026

Covers SIEM deployment, log analysis, incident detection, threat intelligence & SOC workflows

## TECHNICAL SKILLS

|                      |                                                                                                                 |
|----------------------|-----------------------------------------------------------------------------------------------------------------|
| SIEM & SOC:          | Splunk, Log Analysis & Correlation, Incident Response, Threat Hunting, Threat Intelligence, Digital Forensics   |
| Offensive Security:  | Nmap, Metasploit, Burp Suite, Hydra, SQLMap, Gobuster, Nikto, Aircrack-ng, Netcat, LinPEAS, GTFOBins, Responder |
| Reverse Engineering: | Ghidra, x64dbg, PE-bear, YARA, Binary Analysis, PE Header Inspection, Behavioral IOC Extraction, INetSim        |
| Defensive:           | Wireshark, Volatility, OPNsense, OSINT, Vulnerability Assessment, OWASP Top 10, Active Directory, BloodHound    |
| Programming:         | Python, Bash, Go, SQL, HTML/CSS, PowerShell                                                                     |
| Platforms:           | Kali Linux, Parrot OS, REMnux, FlareVM, Windows Server, VirtualBox, Docker, Git & GitHub                        |
| CTF Profiles:        | TryHackMe (@whoopx), HackTheBox (#2902723), PicoCTF (@jishnuec), Blue Team Labs Online, PentesterLab (@whoop)   |

## PROJECTS & KEY ACHIEVEMENTS

### Calicut University — Vulnerability Assessment Report

Web Security, Vulnerability Assessment, Professional Report · 2025

- Situation:** Calicut University's web platform had no prior formal security assessment, leaving unknown attack surfaces exposed.
- Action:** Conducted a structured web vulnerability assessment — mapping endpoints, testing for injection flaws, authentication weaknesses, and misconfigurations using industry-standard methodology.
- Result:** Delivered a professional report with findings categorised by severity, risk ratings, and actionable remediation recommendations — treated as a real-world client deliverable.

### InstaGrep v4.0 — Instagram OSINT Desktop Tool

OSINT, Python, Desktop Application · [github.com/Jishnuec/InstaGrep](https://github.com/Jishnuec/InstaGrep)

- Situation:** Manual Instagram OSINT investigations were slow, inconsistent, and produced no structured output for reporting.
- Action:** Built a ~3,000-line Python desktop application with a full GUI, integrating Instagapi for data extraction, Matplotlib for behavioural pattern charts, and multi-format export (PDF/CSV/JSON).
- Result:** Delivered a complete professional-grade OSINT suite — profile analysis, media download, activity timeline, and report generation — usable in real investigations.

### ReconFrame — Advanced Reconnaissance Framework

Penetration Testing, Python, Tool Development · [github.com/Jishnuec/reconframe](https://github.com/Jishnuec/reconframe)

- Situation:** Penetration test recon required manually invoking 12+ separate tools with no unified workflow or output formatting.
- Action:** Built a Python terminal framework integrating Nmap, Whois, DNS enumeration, and 9+ other tools — with smart IP/domain auto-detection and clean ANSI-formatted output.
- Result:** Single-command recon across all major attack surface dimensions; eliminated manual tool-switching and standardised output for reporting.

### Udyama (उद्यम) — Custom CTF Machine

CTF Design, Linux, Privilege Escalation · Feb 2026 · [archive.org/details/Udyama](https://archive.org/details/Udyama)

- Situation:** The community lacked intermediate-to-advanced Linux privilege escalation CTF challenges grounded in realistic attack techniques.
- Action:** Designed and packaged a custom Ubuntu 22.04 OVA with 2 flags (User + Root), realistic privilege escalation paths, and persistence mechanisms built in.
- Result:** Released publicly on Archive.org — freely available to the global cybersecurity community; demonstrates attack chain design, defender thinking, and deep Linux internals knowledge.

## HOME LAB

Kali vs DVWA — Full Red Team Lab with Splunk SIEM Monitoring

Red Team / Blue Team, SIEM, Web Exploitation, Privilege Escalation, Persistence

|                                                              |                                                                   |                                                                                             |
|--------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| ⌘ ATTACKER<br><b>Kali Linux</b><br>10.0.2.16<br>Network: NAT | 🎯 TARGET<br><b>Linux Mint</b><br>10.0.2.15<br>Apache · SSH · DVWA | ♥ SIEM<br><b>Windows (Splunk)</b><br>Host-Only Adapter<br>Splunk Enterprise · Log Ingestion |
|--------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------|

Network: NAT Network (Kali ↔ Mint) + Host-Only (Mint ↔ Windows / Splunk)

| #    | PHASE        | ATTACK / TECHNIQUE             | TOOL               | OUTCOME                        |
|------|--------------|--------------------------------|--------------------|--------------------------------|
| F-01 | RECON        | Network & Service Enumeration  | nmap -sV -sC -A    | Services & OS Fingerprinted    |
| F-02 | RECON        | Directory & Endpoint Discovery | Dirb + Gobuster    | Hidden Paths Uncovered         |
| F-03 | EXPLOITATION | SSH Credential Brute Force     | Hydra              | Valid Credentials Obtained     |
| F-04 | EXPLOITATION | DVWA Web Login Brute Force     | Hydra              | Authentication Bypassed        |
| F-05 | EXPLOITATION | SQL Injection — Manual + Blind | Manual + SQLmap    | Database Contents Extracted    |
| F-06 | EXPLOITATION | XSS — Reflected & Stored       | Browser + Curl     | Scripts Injected & Confirmed   |
| F-07 | EXPLOITATION | Malicious File Upload + Shell  | Metasploit / PHP   | Reverse Shell Payload Uploaded |
| F-08 | EXPLOITATION | Reverse Shell — Remote Access  | Netcat             | Interactive Shell Established  |
| F-09 | POST-EXPLOIT | Local Privilege Escalation     | LinPEAS + GTFOBins | Root Access Achieved           |
| F-10 | POST-EXPLOIT | Persistence Mechanisms         | Crontab + SSH Keys | Persistent Access Maintained   |

10 Findings Documented · Full Kill Chain · OWASP-style Penetration Test Report

GitHub: github.com/jishnuEC/red-team-soc-dvwa-lab-splunk

Active Directory Red Team Lab — Attack & Hardening

Active Directory, Windows Server 2019, Kerberos Attacks, Post-Exploitation, GPO Hardening

|                                                                                              |                                                                                     |                                                                               |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 🏠 DOMAIN CONTROLLER<br><b>Windows Server 2019</b><br>AD DS · DNS · LDAP<br>Domain: lab.local | 💻 WORKSTATION<br><b>Windows 10</b><br>Domain-joined client<br>Standard user context | ⌘ ATTACKER<br><b>Kali Linux</b><br>Host-Only Network<br>Impacket · BloodHound |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|

- Performed **BloodHound / SharpHound** enumeration to map privilege escalation paths, ACL misconfigurations, and Kerberoastable service accounts across the domain graph.
- Executed **Kerberoasting** and **AS-REP Roasting** attacks using Impacket's GetUserSPNs and GetNPUsers, cracking harvested TGS/AS-REP hashes offline with Hashcat.
- Conducted **Pass-the-Hash** lateral movement via Impacket's psexec and wmiexec, pivoting from a compromised workstation to Domain Admin context.
- Applied targeted hardening post-exploitation: enforced **LAPS**, disabled NTLM where possible, configured fine-grained audit policies, and hardened Kerberos delegation settings via GPO.

Malware Analysis & Reverse Engineering Sandbox

Static Analysis, Dynamic Analysis, Reverse Engineering, Ghidra, x64dbg, YARA, Behavioral IOCs

|                                                                                                   |                                                                                    |                                                                                    |
|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 🔧 ANALYSIS HOST<br><b>FlareVM (Win 10)</b><br>Isolated — no internet<br>Ghidra · x64dbg · PE-bear | 🐞 LINUX TOOLS<br><b>REMnux</b><br>Host-Only Network<br>Strings · YARA · Volatility | 🌐 FAKE INTERNET<br><b>INetSim</b><br>Simulated services<br>DNS · HTTP · SMTP stubs |
|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|

- Conducted **static PE analysis** — inspecting import address tables, section entropy, embedded strings, and packer signatures using PE-bear and FLOSS to triage suspicious binaries before execution.
- Performed **dynamic behavioral analysis** in FlareVM: monitored process creation, registry writes, file-system mutations, and network callbacks using Process Monitor and Wireshark against INetSim.
- Used **Ghidra** for disassembly and code-level reverse engineering — tracing malicious control flow, identifying decryption routines, and reconstructing C2 communication logic.
- Authored **YARA rules** derived from extracted byte patterns and behavioral indicators, enabling signature-based detection of analysed samples and their variants.